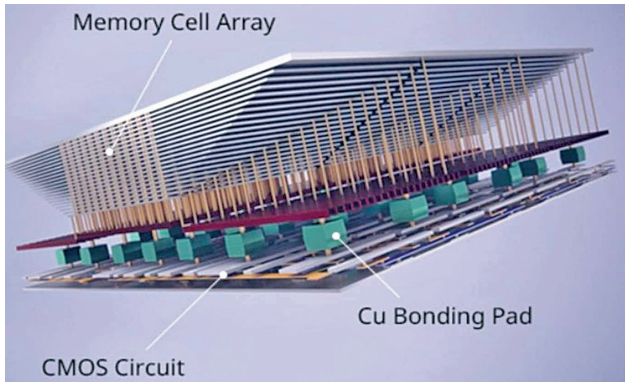


Flash memory can now store more data

Kioxia and Sandisk have unveiled 10th-generation QLC 3D flash memory that increases bit density by up to 60% over their eighth-generation technology, exceeding 37Gb/mm².

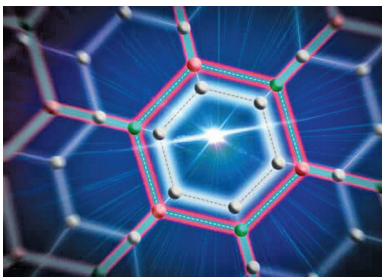


Flash memory can now store more data (Credit: www.electronicshobby.com)

The 332-layer memory uses CMOS directly bonded to array (CBA) architecture and reaches a 4.8Gb/s interface speed with Toggle DDR6.0 and the separate command address protocol. Developed by the companies for AI and cloud infrastructure, the technology improves read/write bandwidth while reducing I/O power consumption and cooling requirements. Its higher density and efficiency could support growing data-storage demands in increasingly data-intensive computing environments.

Ultra-thin superconductors can now stay stable in air

Researchers at MIT have developed a method to produce wafer-scale, air-stable ultrathin superconductors, removing a major barrier to scalable quantum hardware. The technique grows niobium diselenide beneath graphene, which shields the material from oxidation while guiding it into a smooth, continuous monolayer. Engineered by a team in Cambridge, Massachusetts, the process produced films more than 2.5cm wide and enabled integration



The new technique allows researchers to generate an ultrathin superconducting material (represented by pink and green spheres) underneath another atomically thin material—carbon-based graphene (grey) (Image Credit: Jose-Luis Olivares, MIT)

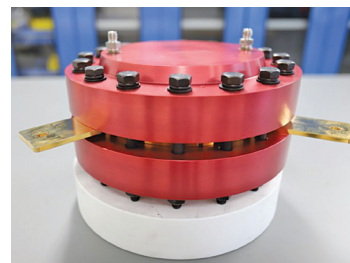
into superconducting microwave circuits. The material retained superconductivity and high kinetic inductance after fabrication. The advance could support smaller quantum computers, ultrasensitive detectors, and practical quantum technologies through reliable fabrication.

GaN technology brings higher voltage for power systems

Power Integrations has raised the voltage rating of its PowiGaN gallium-nitride technology to 2200V, targeting next-generation high-voltage power systems. The technology is designed to support higher-voltage bus architectures while retaining GaN's switching and efficiency advantages. It combines a 2200V-rated GaN switch with an integrated controller and protection features, helping reduce component count and system complexity. The technology is aimed at data centres, electric vehicles, solar power systems, and energy-storage equipment. Developed by Power Integrations, the higher-voltage platform could enable more efficient, compact power-conversion designs as these applications move toward higher operating voltages.

Waste heat can now be used for cooling

Researchers in South Korea have developed a heat pump that converts low-temperature waste heat into cooling, potentially reducing electricity use in factories and data centres. Developed by the Korea Institute of Machinery and



Electrochemical compressor stack (Credit: www.dongascience.com)

Materials with Chung-Ang University, the system operates with waste heat as low as 40°C, compared with more than 70°C required by conventional adsorption cooling systems. Its electrochemical compressor has no moving parts, minimising noise, vibration, and energy consumption, while ammonia serves as a natural refrigerant. A 10kW prototype is undergoing testing for industrial and commercial cooling applications.

Interface tackles nanoscale transistor limits

Researchers from National Yang Ming Chiao Tung University with TSMC Corporate Research in Taiwan have developed an interface-engineering technique that could improve the performance and scalability of atomically thin transistors. The method uses a 0.42-nanometre aluminium oxide layer between monolayer molybdenum disulfide and a hafnium oxide gate dielectric, enabling stronger electrostatic control while preserving carrier transport. The technology produced transistors with roughly 1-nanometre equivalent oxide thickness, low leakage, minimal hysteresis, and high transconductance. The approach could support future low-power logic and advanced semiconductor devices beyond conventional silicon scaling.